

Question Number	Answer	Additional guidance	Mark
4(a)(i)	The only correct answer is C Eukarya <i>A is incorrect because plants are found in the domain Eukarya</i> <i>B is incorrect because plants are found in the domain Eukarya</i> <i>D is incorrect because plants are found in the domain Eukarya</i>		(1)

Question Number	Answer	Additional guidance	Mark
4(a)(ii)	A description that makes reference to the following: • (evidence from) molecular phylogeny (1) • identification of {similarities / differences} in {DNA / RNA / proteins / enzymes / ribosomes / membrane components / cell wall components} (1)	Accept comparing {DNA / RNA / proteins / enzymes / ribosomes / membrane components / cell wall components} Ignore {looking at / analysing} the {DNA / RNA / proteins / enzymes / ribosomes / membrane components / cell wall components / genetic makeup}	(2)

Question Number	Answer	Additional guidance	Mark
4 (b)	An explanation that includes three of the following points: • (tube) grows (through style) to {ovary / ovule / egg cell} / transports the {male nuclei / generative nucleus / male gametes} to {ovary / ovule / egg cell} (1) • (and secretes) digestive enzymes used to aid growth of tube (to ovule) (1) • (therefore) male nuclei can fuse with {egg cell (nucleus) / polar nuclei} (1)	Ignore micropyle Accept forms pathway to reach {ovary / ovule / egg cell} Accept enzymes digest through the style Accept fertilisation occurs to form {zygote / endosperm} Accept double fertilisation occurs / zygote and endosperm formed	(3)

Question Number	Answer	Additional guidance	Mark
4(c)(i)	An answer that makes reference to the following: • as the Verapamil concentration increases the length of the pollen tube decreases (1) • credit a correct manipulated mathematical statement (1)	Accept negative correlation	(2)

Question Number	Answer	Additional guidance	Mark
4(c)(ii)	An answer that makes reference to the following: • lower concentration of calcium ions (in the cells) (1) • fewer {pectin / calcium pectate} molecules formed (1) • therefore increased flexibility of {microfibrils / cellulose} (due to fewer {pectin / calcium pectate} molecules) (1)	Accept no calcium ions present (in the cells) Accept calcium (ions) needed to produce {pectin / calcium pectate} Accept {calcium pectate / pectin} (are needed to) hold {microfibrils / cellulose} together (in a matrix) Accept {microfibrils / cellulose} are not held together by {calcium pectate / pectin} Accept calcium pectate helps middle lamella to hold adjacent cell walls together	(3)